

The Hyperbolic Composition Generator Is Never Bounded and never has an H^∞ calculus on a proper sector

Candidate full result for the Section 3 question in arXiv:1708.02259

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Status: candidate full proof, likely valid. The literal source question has a negative answer for every positive weight sequence. Under the natural alternative reading suggested by the surrounding discussion, every admissible weight sequence gives the requested failure of a proper-sector calculus. Novelty remains subject to expert literature review.

1 Source question and the two readings

Let $\beta = (\beta_n)_{n \geq 0}$ be positive and let

$$\mathcal{H}^2(\beta) = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \sum_{n \geq 0} |a_n|^2 \beta_n^2 < \infty \right\}, \quad \|z^n\| = \beta_n.$$

As in the source, assume that this is a Hilbert space of holomorphic functions satisfying $\mathcal{O}(\overline{\mathbb{D}}) \hookrightarrow \mathcal{H}^2(\beta) \hookrightarrow \text{Hol}(\mathbb{D})$ continuously. Put

$$\alpha_r(z) = \frac{z+r}{1+rz}, \quad \phi_t = \alpha_{\tanh t}, \quad T(t) = C_{\phi_t}.$$

The displayed question on source PDF p. 7 is reproduced below [1].

on $\mathcal{B}$ admits an H^∞ calculus on a half-plane $|\arg(z_0 - z)| < \pi/2$, see the book [18]. The existence of an H^∞ calculus is not affected if one passes to an equivalent norm on $\mathcal{B}$. This motivates the following question.

Question. Do there exist weights $\{\beta_n\}$ such that the generator of the semigroup $\{C_{\alpha_{\tanh t}}\}$ on $\mathcal{H}^2(\beta)$ is bounded, but does not admit an H^∞ calculus in a sector $|\arg(z_0 - z)| < \theta$, where $\theta \in (0, \pi/2)$? The same can be asked for an arbitrary bounded composition semigroup on $\mathcal{H}^2(\beta)$.

Finally, we notice that the property $(\star\star)$ and estimates like (3.2) are known for many classical Banach spaces. We refer to [13, Chapter 3] for H^p spaces and to [9] for VMOA. Property $(\star\star)$ also holds true for the case of mixed norm spaces $H(p, q, \alpha)$, see [2].

Exact transcription. “Do there exist weights $\{\beta_n\}$ such that the generator of the semigroup $\{C_{\alpha_{\tanh t}}\}$ on $\mathcal{H}^2(\beta)$ is bounded, but does not admit an H^∞ calculus in a sector $|\arg(z_0 - z)| < \theta$, where $\theta \in (0, \pi/2)$? The same can be asked for an arbitrary bounded composition semigroup on $\mathcal{H}^2(\beta)$.”

Taken literally, the generator is required to be a bounded operator. The surrounding paragraphs instead contrast bounded composition operators with quasicontractivity, so a plausible intended reading is that the *semigroup consists of bounded operators* while its generator lacks the proper-sector calculus. Theorem 4 resolves both readings. Here “bounded composition semigroup” is used in the source’s pointwise sense: each $T(t)$ is a bounded operator, not necessarily $\sup_{t \geq 0} \|T(t)\| < \infty$.

Proof Intuition

The weight sequence can make one of the two shifts in $(1 - z^2)d/dz$ small, but not both. On two adjacent normalized monomials, the relevant shift coefficients are reciprocal up to the factor $n(n+1)$; their product therefore diverges independently of the weights. This rules out a bounded generator.

For the functional-calculus statement, the hyperbolic maps form a group, not merely a one-sided flow. Reflection in the origin turns every positive-time composition operator into its negative-time counterpart. A proper-sector H^∞ calculus would make the semigroup analytic. But an analytic semigroup that also extends to a C_0 -group has bounded generator, because positive time maps the whole space into the generator domain and negative time inverts that map. The monomial obstruction then gives the contradiction.

2 The weight-independent monomial obstruction

The derivative of the flow at zero is

$$\left. \frac{d}{dt} \phi_t(z) \right|_{t=0} = 1 - z^2.$$

Under the hypotheses above, the source's generator-domain formula gives

$$Af = (1 - z^2)f', \quad \mathcal{D}(A) = \{f \in \mathcal{H}^2(\beta) : (1 - z^2)f' \in \mathcal{H}^2(\beta)\}. \quad (1)$$

In particular every polynomial belongs to $\mathcal{D}(A)$.

Lemma 1 (Adjacent-monomial obstruction). *For every positive sequence β , the operator in (1) is unbounded on $\mathcal{H}^2(\beta)$.*

Proof. Let $e_n = z^n / \beta_n$. These are unit vectors, and for $n \geq 1$,

$$Ae_n = n \frac{\beta_{n-1}}{\beta_n} e_{n-1} - n \frac{\beta_{n+1}}{\beta_n} e_{n+1}. \quad (2)$$

Orthogonality gives

$$\|Ae_n\| \geq n \frac{\beta_{n+1}}{\beta_n}, \quad \|Ae_{n+1}\| \geq (n+1) \frac{\beta_n}{\beta_{n+1}}.$$

The product of the right-hand sides is $n(n+1)$. Hence

$$\max\{\|Ae_n\|, \|Ae_{n+1}\|\} \geq \sqrt{n(n+1)} \longrightarrow \infty. \quad (3)$$

Thus A is unbounded on its domain, for every choice of positive weights. $\square$

The use of adjacent inputs is essential. Looking only at Ae_n permits the ratio β_{n+1}/β_n to suppress its forward coefficient; the backward coefficient on Ae_{n+1} is then forced to grow by the reciprocal amount.

3 The group and proper-sector obstruction

Lemma 2 (The positive flow automatically extends to a group). *Suppose $T(t) = C_{\phi_t}$, $t \geq 0$, is a C_0 -semigroup on $\mathcal{H}^2(\beta)$. Then the same formula for $t \in \mathbb{R}$ defines a C_0 -group.*

Proof. Let $\rho(z) = -z$ and $R = C_\rho$. Since the monomials are orthogonal, R is unitary on $\mathcal{H}^2(\beta)$. Direct calculation gives

$$\rho \circ \alpha_r \circ \rho = \alpha_{-r}, \quad C_{\alpha_{-r}} = RC_{\alpha_r}R.$$

Thus all negative-time operators are bounded and are conjugates by the fixed unitary R . Strong continuity for negative times follows from strong continuity for positive times, and $\phi_{t+s} = \phi_t \circ \phi_s$ gives the group law. $\square$

Lemma 3 (An analytic C_0 -group has bounded generator). *If a C_0 -group on a Banach space is analytic for positive time, then its generator is bounded.*

Proof. Fix $s > 0$. Analyticity implies $T(s)X \subset \mathcal{D}(A)$ and $AT(s) \in \mathcal{L}(X)$. Since $T(-s)$ is bounded and $T(s)T(-s) = I$, every $x \in X$ belongs to $\mathcal{D}(A)$ and

$$A = (AT(s))T(-s) \in \mathcal{L}(X).$$

$\square$

For $z_0 \in \mathbb{C}$ and $0 < \theta < \pi$, write

$$S_\theta(z_0) = \{z_0 - w : w \neq 0, |\arg w| < \theta\}.$$

We use the standard meaning of a bounded H^∞ calculus on this shifted left sector. Such a calculus for A makes $B = z_0 - A$ sectorial of angle at most θ . If $\theta < \pi/2$, the sectorial generation theorem makes e^{-tB} analytic, and therefore makes $T(t) = e^{tz_0} e^{-tB}$ analytic as well [2].

Theorem 4 (Full resolution for the hyperbolic flow). *Let β be any positive weight sequence satisfying the source hypotheses, and suppose $T(t) = C_{\alpha_{\tanh t}}$ is a C_0 -semigroup on $\mathcal{H}^2(\beta)$. Its generator A has the following properties.*

1. A is unbounded.

2. For no $z_0 \in \mathbb{C}$ and no $0 < \theta < \pi/2$ does A admit a bounded H^∞ calculus on $S_\theta(z_0)$.

Proof. Part (1) is Lemma 1. If the calculus in (2) existed, the preceding sectorial generation argument would make T analytic. The group extension and the analytic-group lemma would then make A bounded, contradicting (1). $\square$

Corollary 5 (Answers to the two readings). *The literal existence question has answer no: the stated generator is never bounded. If “bounded” was intended to modify the composition semigroup, in the source’s sense that every member is bounded, then every admissible choice of weights has the asserted failure of all proper-sector calculi. In particular the classical choice $\beta_n = 1$ is an example.*

This does not contradict the half-plane statement immediately preceding the question: the boundary angle $\pi/2$ is allowed there, while the theorem excludes exactly the strict range $0 < \theta < \pi/2$.

4 Verification, limitations, and novelty

The accompanying verifier checks the flow and reflection identities and the adjacent-monomial lower bound for constant, alternating, polynomial, geometric, and pseudorandom positive weights. The checks pass through $n = 500$; they are sanity checks for the exact algebraic proof, not a substitute for it.

Adversarial checks. No regularity of the ratios β_{n+1}/β_n is assumed. The proof uses two adjacent normalized monomials precisely to prevent extreme ratios from evading the estimate. Polynomials belong to the generator domain by (1). The proper-sector conclusion uses bounded H^∞ calculus in the standard sectorial sense; a half-plane calculus at angle $\pi/2$ remains possible. The group argument does not assume uniform boundedness in time.

Scope limitation. The theorem fully resolves the fixed hyperbolic-flow question, including its wording ambiguity. It does not classify H^∞ calculi for arbitrary noninvertible composition semiflows, nor does it claim that every composition-semigroup generator is unbounded.

Bounded novelty audit. The four run indexes and focused primary-source searches through 13 August 2026 were checked using arXiv:1708.02259 and combinations of “weighted Hardy”, “hyperbolic composition”, “generator”, “sectorial”, and “ H^∞ calculus”. No later paper explicitly answering the displayed question was found. Novelty is plausible, not certified.

Human-review recommendation: send to an operator-semigroup or composition-operator expert. The highest-value review points are the interpretation of “bounded” in the source wording and the standard implication from a proper-sector bounded H^∞ calculus to analyticity.

References

- [1] E. A. Gallardo-Gutiérrez and D. V. Yakubovich, *On generators of C_0 -semigroups of composition operators*, Israel J. Math. **229** (2019), 487–500; arXiv:1708.02259.
- [2] M. Haase, *The Functional Calculus for Sectorial Operators*, Operator Theory: Advances and Applications 169, Birkhäuser, 2006.